

Olist E-Commerce Operations Playbook

Action Plan Recommendations for Delivery, Review, Pareto, and Target-Gap Diagnostics

(Generated for a public Databricks Genie and AI/BI portfolio demo using the Olist Brazilian E-Commerce dataset.)

1) KPI Reference and How to Interpret Shifts

This playbook converts Genie diagnostics into operational recommendations for category performance, delivery reliability, customer review quality, and Pareto-based prioritization.

KPIs covered

- `total_order_value`: Total merchandise/order value, excluding freight.
- `order_count`: Distinct order count.
- `late_delivery_rate`: Share of delivered orders arriving after estimated delivery.
- `average_review_score`: Average customer review score for reviewed orders.
- `review_score_gap_late_vs_on_time`: Observed review score gap between on-time and late delivered orders.
- `late_delivery_rate_reduction_gap_to_4_2_review`: Modeled reduction in late delivery rate needed to reach a 4.2 review score when feasible.

General interpretation rules

- Always prioritize issues using both operational risk and business contribution.
- Use Pareto buckets to distinguish high-impact categories or states from long-tail noise.

- Treat target-gap fields as modeled what-if estimates, not causal proof.
- If on-time review scores are below 4.2, late-delivery reduction alone is insufficient.
- For review questions, check review coverage and avoid overinterpreting tiny segments.
- Grain context: See KNOWLEDGE_BASE.md Source Table section..
- Common filters include order_id, customer_id, order_status, customer_city, customer_state, seller_city, seller_state, product_category.

2) Scenario Playbooks (Symptoms → Likely Causes → Action Plans)

Scenario A — High-Value Category With Elevated Late Delivery

Symptoms

- Product category has order_value_share above 1% and late_delivery_rate above 8%.
- Category appears in the Pareto top 80 percent of order value.

Likely causes

- Fulfillment constraints in high-volume categories.
- Seller capacity mismatch for popular products.
- Freight network delays concentrated in a few seller or customer states.

Action plan recommendations

1. Immediate actions (today-48h)

- Rank affected categories by order_value_share and late_delivery_rate.
- Split the category by seller_state and customer_state to isolate concentration.

- Check whether late orders also show lower review scores.

2. Short-term actions (1-4 weeks)

- Create seller outreach for high-value late-delivery categories.
- Review carrier handoff and shipping-limit patterns for the category.
- Monitor daily `late_delivery_rate` and `review_score_gap_late_vs_on_time`.

Measurement targets

- Reduce `late_delivery_rate` for prioritized categories by 10-20% relative.
- Improve `average_review_score` or reduce `review_score_gap_late_vs_on_time`.

Scenario B — Late Delivery Strongly Affects Review Score

Symptoms

- `review_score_gap_late_vs_on_time` is among the largest category gaps.
- `avg_review_when_late` is materially lower than `avg_review_when_on_time`.

Likely causes

- Customer expectations are highly delivery-sensitive in this category.
- Long delivery times create dissatisfaction independent of product quality.
- Communication gaps around delayed delivery.

Action plan recommendations

1. Immediate actions (today-48h)

- List categories by `review_score_gap_late_vs_on_time` descending.
- Separate categories where `average_review_score` is already below target.
- Inspect whether late deliveries are concentrated in specific customer states.

2. Short-term actions (1-4 weeks)

- Add proactive delay messaging for sensitive categories.

- Set delivery reliability targets for sellers in high-gap categories.
- Track avg_review_when_late and avg_review_when_on_time separately.

Measurement targets

- Narrow review_score_gap_late_vs_on_time.
- Increase avg_review_when_late without degrading on-time order review scores.

Scenario C — Review Target Gap To 4.2

Symptoms

- late_delivery_rate_reduction_gap_to_4_2_review is positive and large.
- review_4_2_target_status is late_delivery_reduction_can_help.

Likely causes

- Late delivery is a material lever for review improvement.
- A small number of categories may need disproportionate delivery improvement.
- Current late delivery rate exceeds modeled target rate.

Action plan recommendations

1. Immediate actions (today-48h)

- Rank categories by late_delivery_rate_reduction_gap_to_4_2_review.
- Exclude categories where review_4_2_target_status says late delivery alone is insufficient.
- Pair each category with order_count to avoid overfitting small samples.

2. Short-term actions (1-4 weeks)

- Set category-specific late delivery thresholds.
- Review seller/carrier performance for the target-gap categories.
- Create dashboard alerts for categories drifting away from required late delivery rate.

Measurement targets

- Move late_delivery_rate toward late_delivery_rate_needed_for_4_2_review.
- Increase average_review_score toward 4.2 while maintaining order volume.

Scenario D — Customer-State Pareto Concentration

Symptoms

- A small set of customer states drives the Pareto top 80 percent of orders.
- One of those states also has low average_review_score or high late_delivery_rate.

Likely causes

- Geographic demand concentration amplifies localized delivery issues.
- Carrier performance varies by destination state.
- Regional customer expectations differ.

Action plan recommendations

1. Immediate actions (today-48h)

- Rank customer states by order_count_share and cumulative_order_count_share.
- Overlay late_delivery_rate and average_review_score for Pareto states.
- Identify whether issues are customer-state or seller-state driven.

2. Short-term actions (1-4 weeks)

- Prioritize operational review for Pareto states with poor review or delivery metrics.
- Create region-specific delivery improvement targets.
- Track state-level metrics in the AI/BI dashboard.

Measurement targets

- Improve late_delivery_rate in Pareto states.

- Lift average_review_score for high-volume low-score states.

3) Recommendation Templates (Vector-Friendly Snippets)

These are short, reusable action cards designed for embedding and retrieval.

Action Card: Pareto Category Triage

Trigger: A category is in pareto_top_80_value and has elevated late delivery.

Actions:

- Rank by order_value_share, then by late_delivery_rate.
- Check review_score_gap_late_vs_on_time before prioritizing.
- Create category-specific owner and target.

Expected impact: Focuses operations on segments with both business impact and customer risk.

Action Card: 4.2 Review Target Feasibility Check

Trigger: A stakeholder asks how to reach a 4.2 average review score.

Actions:

- Use review_4_2_target_status to determine feasibility.
- If feasible, use late_delivery_rate_reduction_gap_to_4_2_review.
- If infeasible, add non-delivery CX interventions.

Expected impact: Prevents overclaiming what delivery improvements alone can accomplish.

Action Card: Late Delivery Review Gap Drilldown

Trigger: review_score_gap_late_vs_on_time is high.

Actions:

- Compare avg_review_when_late and avg_review_when_on_time.

- Filter to categories with adequate order_count.
- Prioritize categories where late_delivery_rate is also high.

Expected impact: Identifies where delivery reliability most visibly affects customer satisfaction.

4) Putting It Together: Example "Recommended Plan" Outputs

Example Plan 1 (Audio category has high late delivery and low late-order reviews)

Observed trend: Audio appears with elevated late_delivery_rate and a large review_score_gap_late_vs_on_time.

Plan:

- Validate late delivery concentration by seller_state and customer_state.
- Create a seller/carrier follow-up list for delayed audio orders.
- Track late_delivery_rate, avg_review_when_late, and average_review_score weekly.

Success metrics: Late delivery rate falls, average late-order review score improves, and review score gap narrows.